

פקולטה להנדסת תוכנה

בדיקת מערכות ספרתיות ואימות תכנון

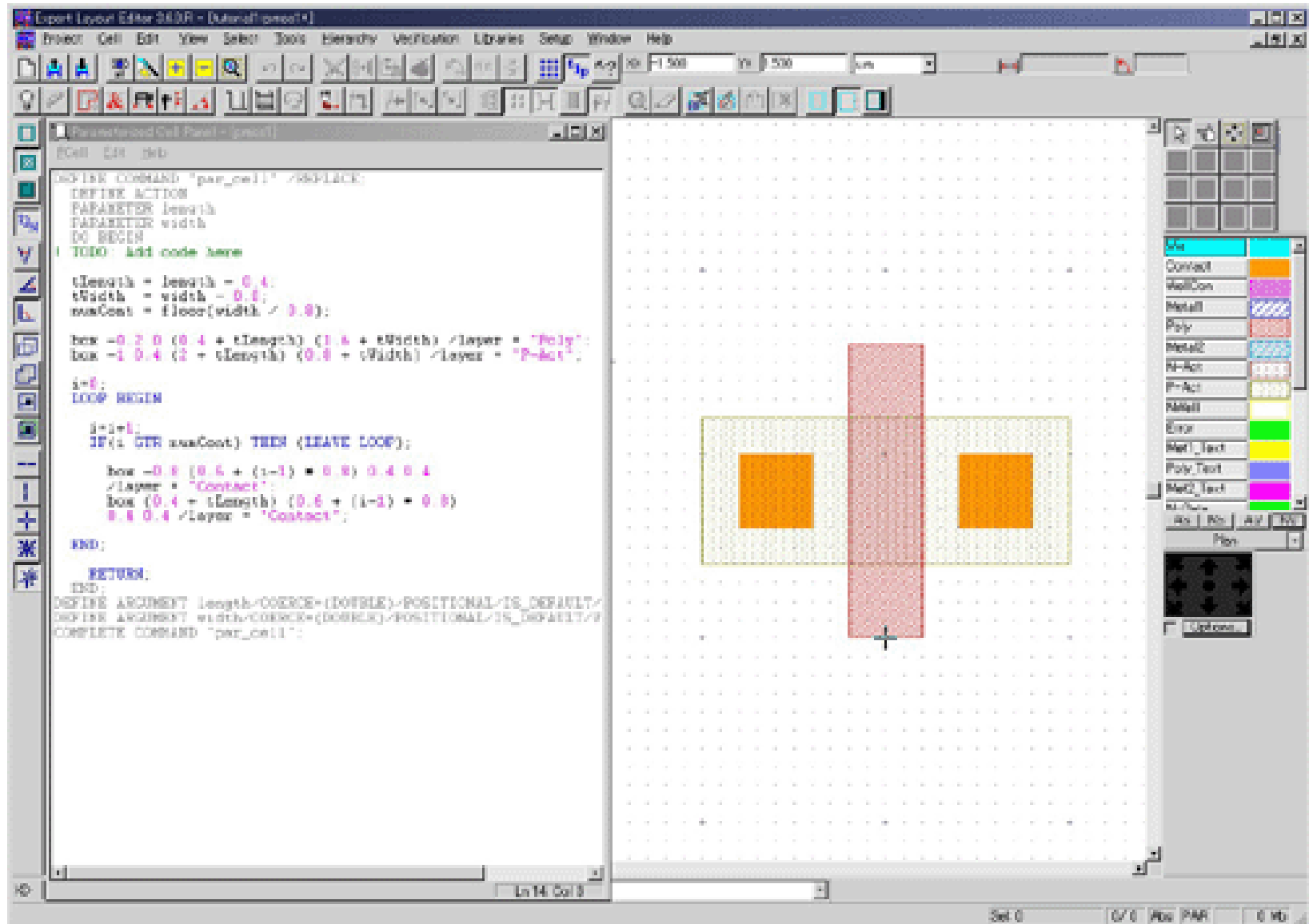
פרק 6

PDK

3.5

3.5

Parameterized Cells



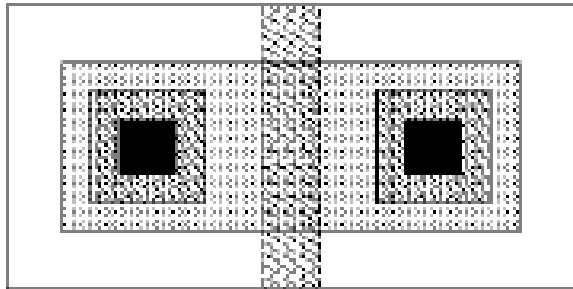
Parameterized cell

- לעתים קרובות במהלך תכנון מעגלים, יש צורך לערוך מופעים רבים של תא בודד (למשל מהפך) כאך שלכל מופע יכולים להיות פרמטרים שונים (למשל, בגדלים שונים טרנזיסטור)
- דרך אחת שבה זה יכול להתממש היא פשוט לייצר מופעי טרנזיסטורים מרובים כל אחד בגודל שלו
- פתרון טוב יותר הוא ליצור תא פרמטרי (PCell)

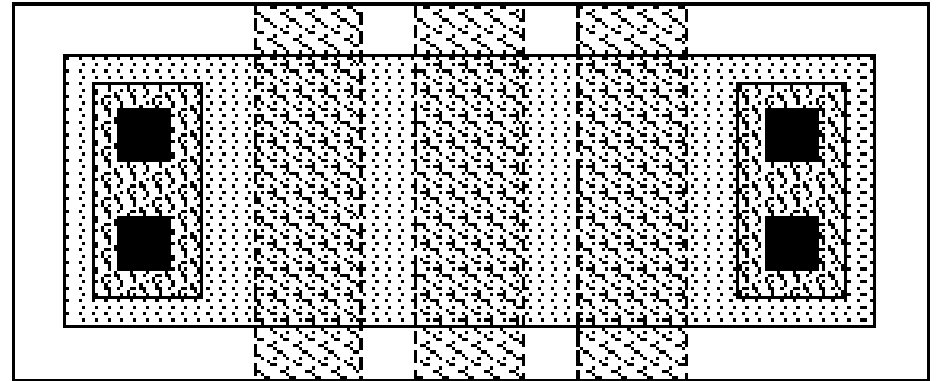
Parameterized cell

- **PCell** קיצור עבור תא פרמטרי, מושג בשימוש נרחב בתכנון אוטומטי אנלוגי (או יגיטלי מותאם אישית) של מעגלים אלקטרוניים
- **PCell** הוא תא אשר נוצר באופן אוטומטי על ידי תוכנות אוטומציה האלקטרונית (EDA electronic design automation)
- התכנון מבוסס על הערכים של הפרמטרים שלו
- **PCell** מייצג חלק או רכיב של מעגל אשר תלוי באחד או יותר פרמטרים, ז"א אינו רכיב קבוע

התא לאחר הפרמטרים השתנו



Original layout



Instance of pcell with different values for gate, width, and length parameters

Parameterized cell (ctnd)

- תא נתון עשוי להיות "ממוקם" (instantiated) פעמים רבות
- PCell אחד יכול לקחת מידות של טרנזיסטור (רוחב ואורך) כפרמטרים
- המופעים (instances) שונים של pcell אחד יכול לייצג טרנזיסטורים בגדלים שונים, אך מאפיינים דומים אחרים

Parameterized cell (ctnd)

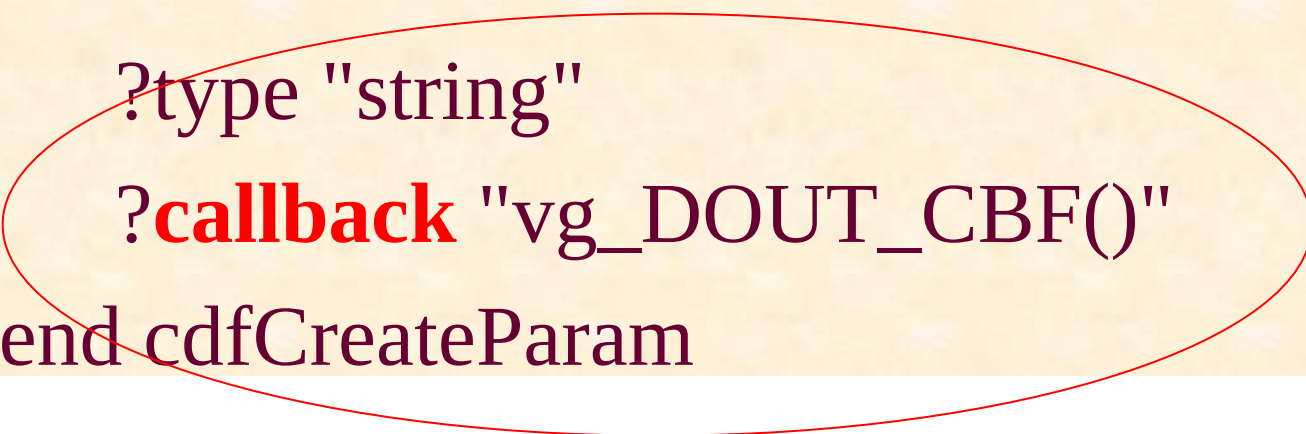
- **Pcell** היא חתיכת קוד
- קוד זה אחראי על תהליך יצירת המבנה התקין של גרסאות **PCell** על פי פרמטרים (קלט) שלו
- חתיכה אחת של קוד יכולה ליצור אובייקטים שונים (עם ערכי הפרמטרים שונים), היא נקראת מאסטר **Pcell**
- **SKILL** ו-**Python** משמשים לעתים קרובות ביותר לכתיבת קוד של **Pcell**. **SKILL** מבוסס על השפה הפופולרית של בינה מלאכותית **Lisp**

Instance of a Pcell

- שינויים של הפרמטרים מתבצעים במאסטר בזמן הקומפילציה של המאסטר
- למשל, ניתן ליצור טרנזיסטור עם פרמטרים שונים אשר בשליטה: אורך, רוחב, ומספר השערים

Component Description Format (CDF)

```
cdfCreateParam(cdfid  
    ?name "dout"  
    ?prompt "Outer Dimension"  
    ?parseAsNumber "yes"  
    ?units "lengthMetric"  
    ?defValue "200u"  
    ?type "string"  
    ?callback "vg_DOUT_CBF()  
);end cdfCreateParam
```



Callback functions

- פונקציות Callback מחוברות לכל CDF פרמטר
- כל הגבלות ותלויות הפרמטרים מקודדות שם

לדוגמא:

```
when(evalstring(dout~>value) < (evalstring(w~>value)+evalstring(s~>value))*2
;incorrect W value
printf("\n*WARNING* Inductor Pcell: Incorrect value for parameter \"Metal
Width\" - %s\n Value will be reset to the minValue - %s" w~>value
pcExprToString((evalstring(w~>value)+evalstring(s~>value))*2))
;set W value to the min allowed
dout~>value = pcExprToString((evalstring(w~>value)+evalstring(s~>value))*2)
```

Coding the p-cell

הפונקציה מחשבת את הקואורדינטות של נקודות
פינתיות של סליל

```
procedure(vgCalculateIndPathPoints(dout w n s)
prog((points p1 p2 p3 p4 p5 offset point X Y)
p1 = 0:0
p2 = 0:dout-w/2+30
p3 = dout-w:cadr(p2)
p4 = car(p3):w/2+30
p5 = w+s:w/2+30
points = list(p1 p2 p3 p4 p5)
;number of points is 4*n+1
```

Coding the p-cell (ctnd)

```
when(n>1
  for(i 6 int(4*n+1)
    point = list(0 0) ; current point
    if(i == int(4*n+1) then
      ;current point is the last point
      offset = w/2
    else
      ;current point is not the last one
      offset = 0
    );end if
```

Coding the p-cell (ctnd)

```
if(evenp(i) then
```

```
  ;i is even
```

```
  ;i point's X coordinate is the same as the i-1 point
```

```
    if(int(evalstring(sprintf(nil "%n.0/4" i)) )!=  
      float(evalstring(sprintf(nil "%n.0/4" i)) ) then
```

```
      ;going UP
```

```
      Y = cadr(nth(i-4 points))-w-s
```

```
      point = list(car(nth(i-2 points)) Y)
```

```
    else
```

```
      ;going DOWN
```

Coding the p-cell (ctnd)

;going DOWN

Y = cadr(nth(i-4 points))+w+s

point = list(car(nth(i-2 points)) Y)

);end if going up or down

else

Coding the p-cell (ctnd)

```
;i is odd  
;i point's Y coordinate is the same as the i-1 point  
if(int(evalstring(sprintf(nil "%n.0/4" i-1)) )!=  
float(evalstring(sprintf(nil  
"%n.0/4" i-1)) ) then  
;going RIGHT  
X = car(nth(i-4 points))-w-s+offset  
point = list(X cadr(nth(i-2 points)))  
else
```

Coding the p-cell (ctnd)

```
;going LEFT
```

```
X = car(nth(i-4 points))+w+s-offset
```

```
point = list(X cadr(nth(i-2 points)))
```

```
);end if going up or down
```

```
);end if evenp(i)
```

Coding the p-cell (ctnd)

```
        points = append1(points point)
    );end foreach point excluding the first five
);end when n>1
if(n==1 then
    ;Underpass not needed
    points = append1(points list(car(nth(4 points)) 0))
);end if n=1
return(points)
);end prog
);end procedure
```

דוגמא: פרמטרים של MOS

Parameters	Choices / default value	Usage
w	Minimum technology value	Width of the gate (in meter)
l	Minimum technology value	Length of the gate (in meter)
m	1	Multiplication factor
splitting	True / False	Specified mos splitting
Fingers	2	Number of fingers in case of mos splitting
Left Contact	True / False	Specify the existence of contact on left side (for abutment)
Right Contact	True / False	Specify the existence of contact on right side (for abutment)
Left Stretch	Technology value	Specify the extension of active over poly when left contact are not present
Right Stretch	Technology value	Specify the extension of active over poly when right contact are not present

דוגמא: פרמטרים של MOS

Source Contact Coverage	100	Specify the percentage of active area (source) covered by contact (default value is the max value)
Drain Contact Coverage	100	Specify the percentage of active area (drain) covered by contact (default value is the max value)
Source Contact Position	Top / Center / Bottom	Alignment of contact on source when source contact coverage is lower than 100
Drain Contact Position	Top / Center / Bottom	Alignment of contact on source when source contact coverage is lower than 100
Add substrate contact?	True / False	Add the bulk
Substrate Contact Position	Top / Bottom / Right / Left	Specify the position of the bulk
Add substrate well (for P device)	True / False	Add N-well layer around the device
Internally connect poly?	True / False	Connect all the gates together in case of mos splitting
Connect poly	Top / Bottom	Specify the position of the gate connection

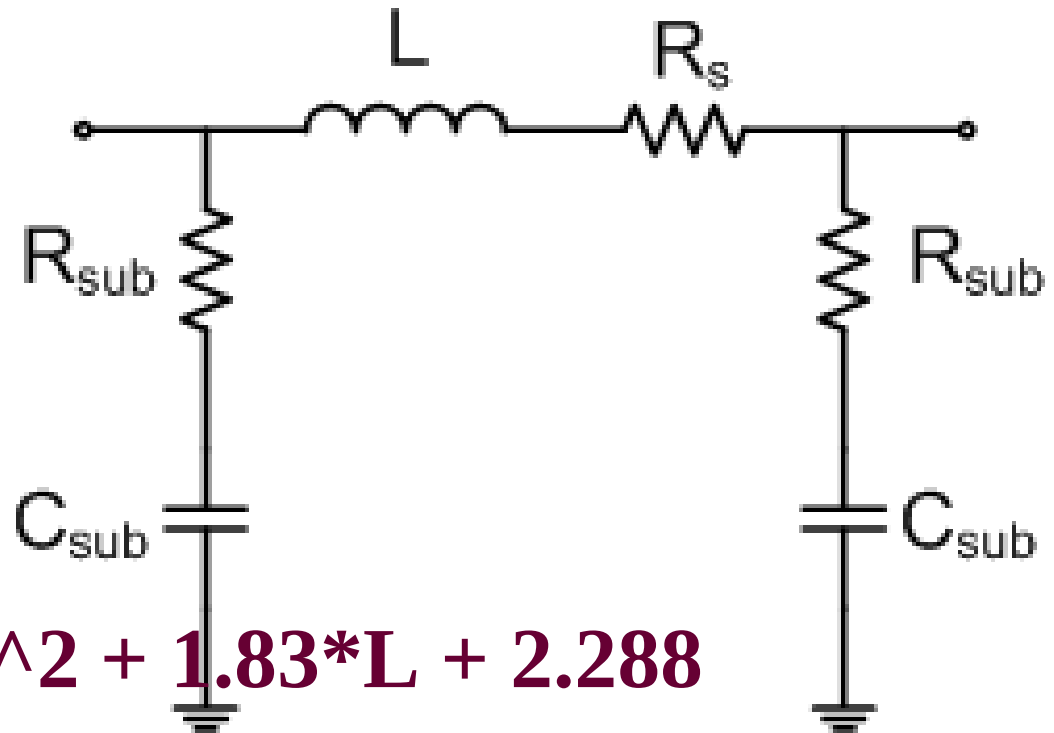
דוגמא: פרמטרים של MOS

Connect S/D to metal?	None / Source / Drain / Source&Drain	Connect Source or Drain together in case of mos splitting
Model name	MOS MOSOX	Model name for simulator
Drain diffusion area	Non editable Computed automatically from w	as, ad, ps, pd value are used by simulator (eldo, spectre ...)
Source diffusion area		
Drain diffusion periphery		
Source diffusion periphery		

דוגמא: פרמטרים של נגד

parameters	Choices / default value	Usage
Resistance	r_min	Value of the resistance
Length	Not editable Computed automatically from R and W	Length of one segment of the resistance
Width	w_min	Width of one segment of the resistance
Number of segment	1	Number of segments
Form	Serial / parallel	Kind of the resistor (parallel or series segment)
Simulation Model	RPOLYHIGH	Name of the model
SheetRes	Not editable	Value of the sheet resistance (Ω/sq)
SheetRes_head	Not editable	Value of the sheet resistance for the head of the resistance ($\Omega/\mu\text{m}$)
dw	Not editable	Width imprecision
r_min	Not editable	Minimum resistance value allowed
w_min	Not editable	Minimum width value allowed
l_min	Not editable	Minimum length value allowed
segsimu	Not editable	Number of segment for simulation (eldo, spectre...)

רכיב של סליל: תכנון האררכי



where:

- $R_s = -0.0296 * L^2 + 1.83 * L + 2.288$
- $C_{sub} = -0.0005 * L^2 + 0.0312 * L + 0.0543$
- $R_{sub} = 0.0789 * L + 31.707 + 3.4892 / L$

Schematic cell view

The image shows a screenshot of the Schematic Editor L Editing: LabProj interface. The main window displays a schematic diagram with a grid background. A component, labeled 'L0', is highlighted with a dashed orange border. The component is an inductor, represented by a green coil symbol. The schematic also includes a capacitor labeled 'C0' with a value of 'c:1p' and a resistor labeled 'R0' with a value of 'r:1e-9'.

The 'Edit Object Properties' dialog box is open, showing the following settings:

- Apply To:** only current (selected), instance (selected)
- Show:** ☐ system, ☒ user, ☒ CDF
- Buttons:** Browse, Reset Instance Labels Display
- Property Value Display:**

Property	Value	Display
Library Name	analogLib	off
Cell Name	ind	off
View Name	symbol	off
Instance Name	L0	off

- Buttons:** Add, Delete, Modify
- CDF Parameter Value Display:**

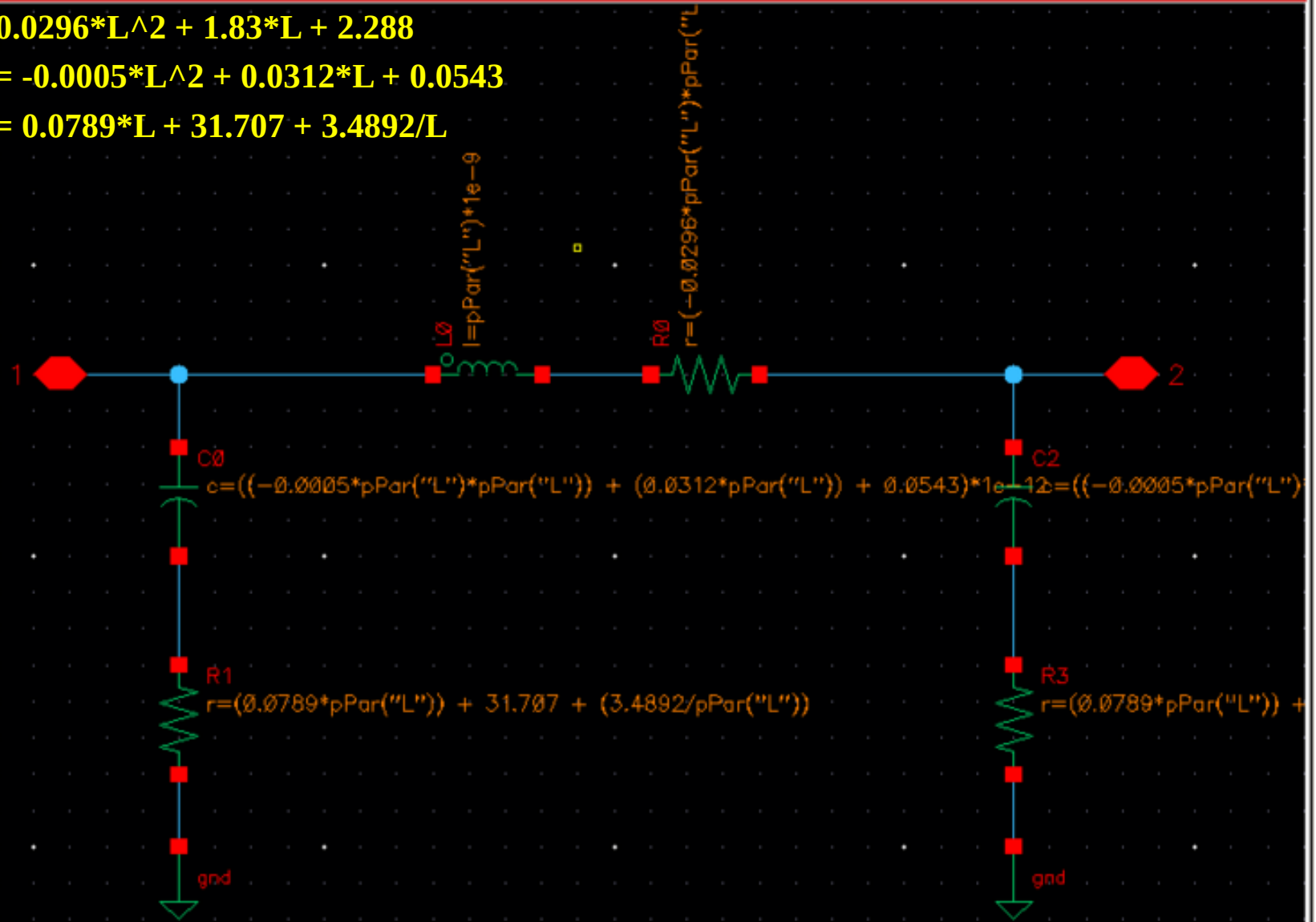
CDF Parameter	Value	Display
Model name		off
Inductance	pPar("L")*1e-9 H	off
Resistance		off
Multiplier		off
Temp rise from ambient		off
Initial condition		off
Linear temp. coefficient		off
Quadratic temp. coeff.		off
Generate noise?	☒	off
Scale factor		off
Number of Polynomial Coeffs	0	off

Schematic cell view

$$R_s = -0.0296*L^2 + 1.83*L + 2.288$$

$$C_{sub} = -0.0005*L^2 + 0.0312*L + 0.0543$$

$$R_{sub} = 0.0789*L + 31.707 + 3.4892/L$$



מאפיינים של הסליל

Edit Object Properties

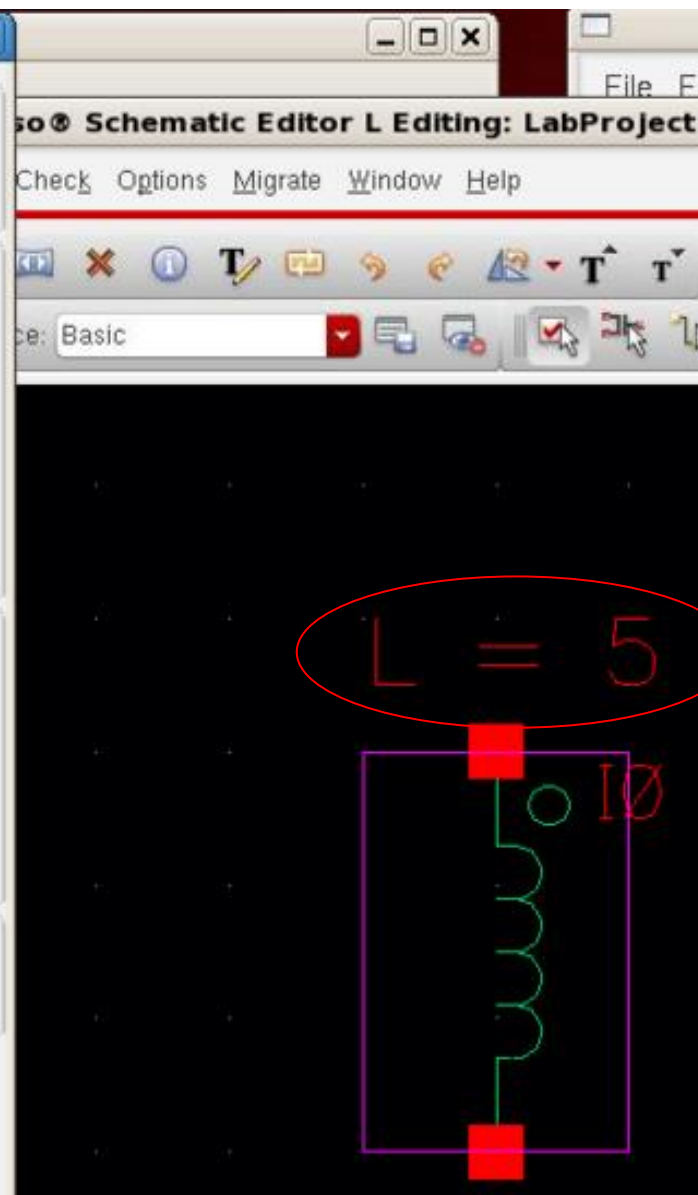
Apply To:

Show: ☐ system ☒ user ☒ CDF

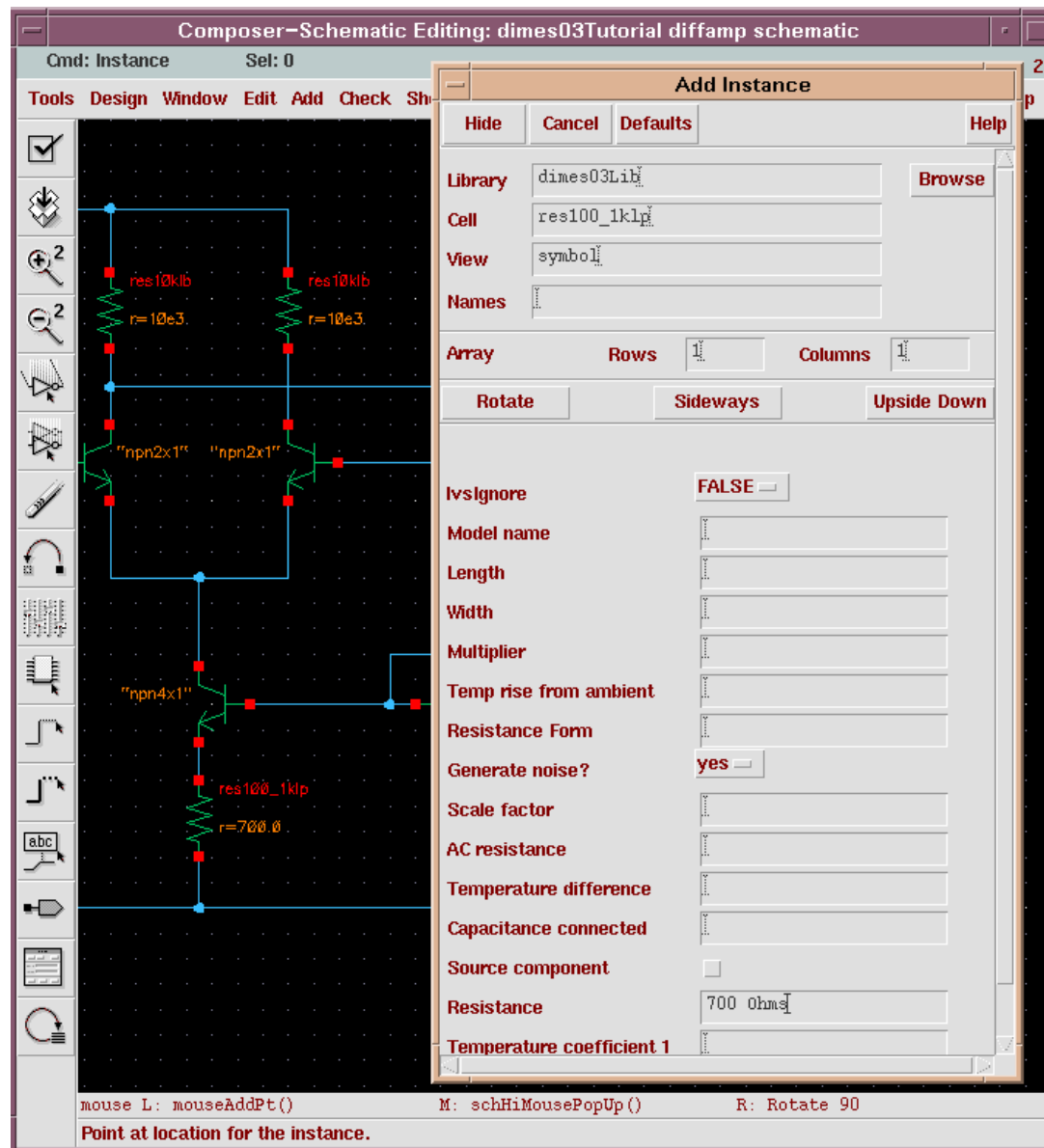
Property	Value	Display
Library Name	LabProject	off
Cell Name	ind_real	off
View Name	symbol	off
Instance Name	I0	off

User Property	Master Value	Local Value	Display
interfaceLastCh...	1 00:16:24 2011		off
partName	ind_real		off
vendorName			off

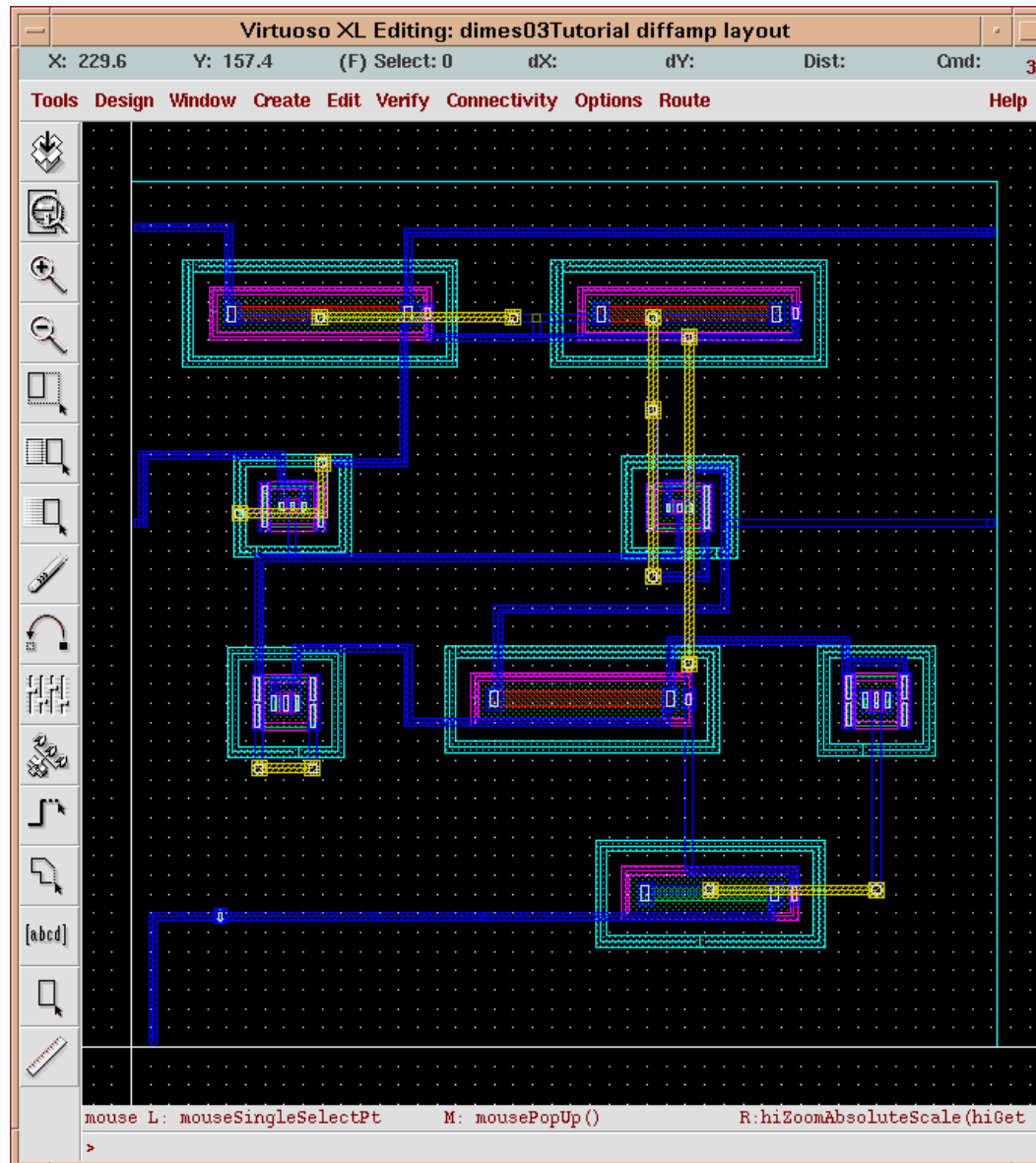
CDF Parameter	Value	Display
L	5	both



רכיב של נגד: schematic

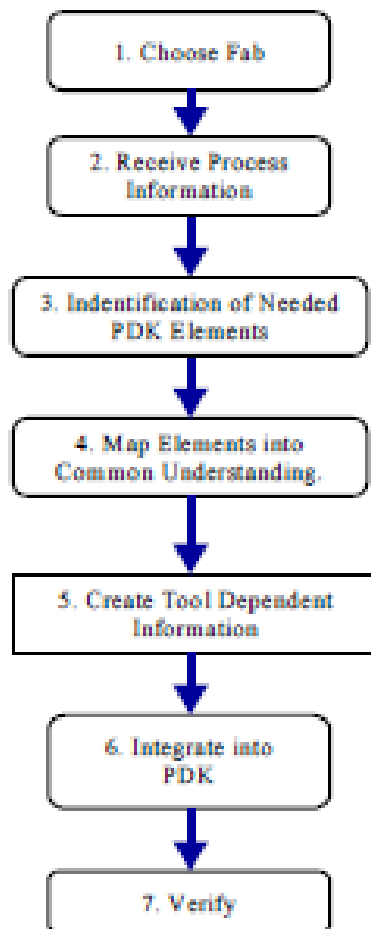


רכיב של נגד: layout

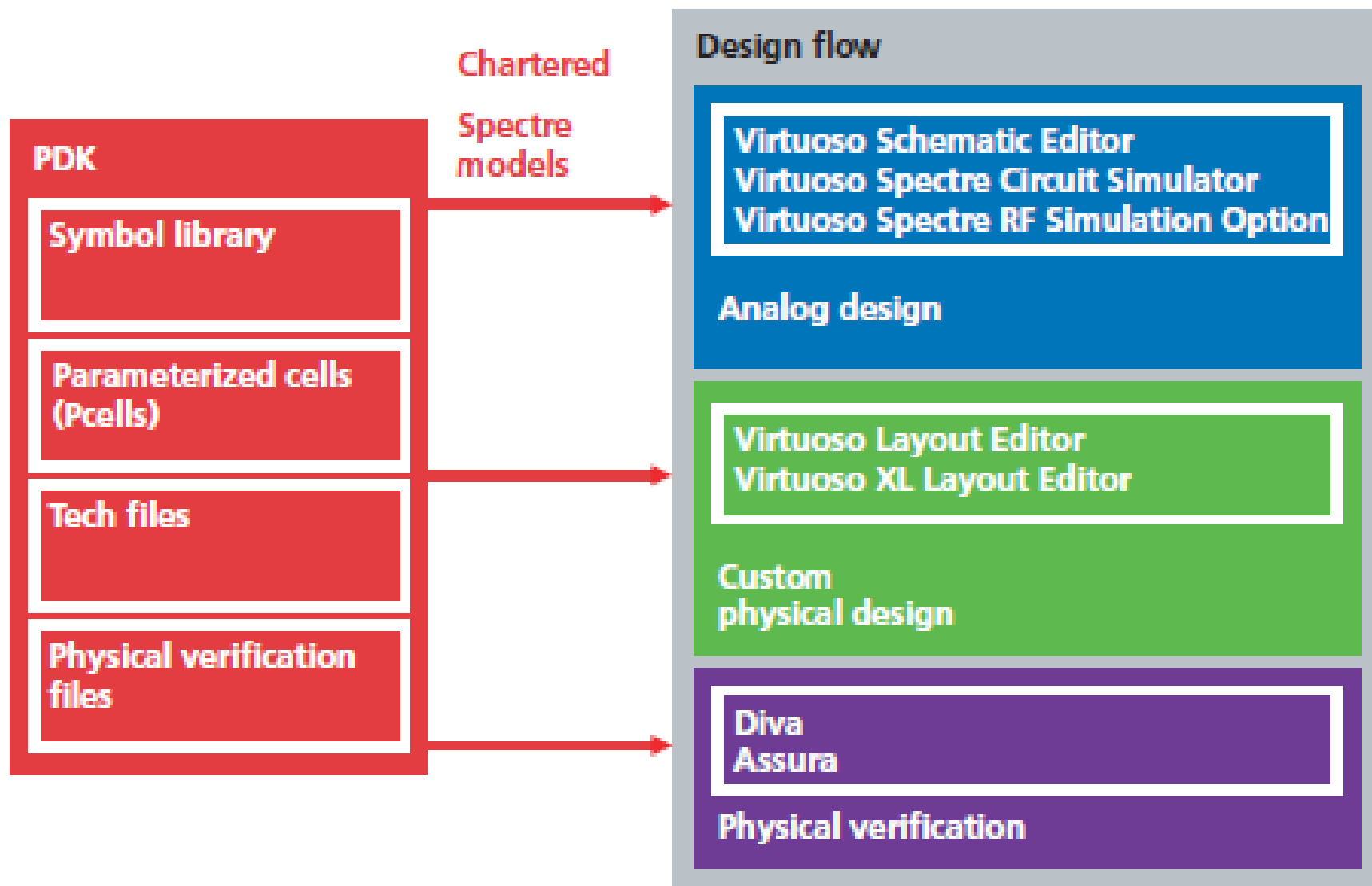


Physical Design Kit (PDK)

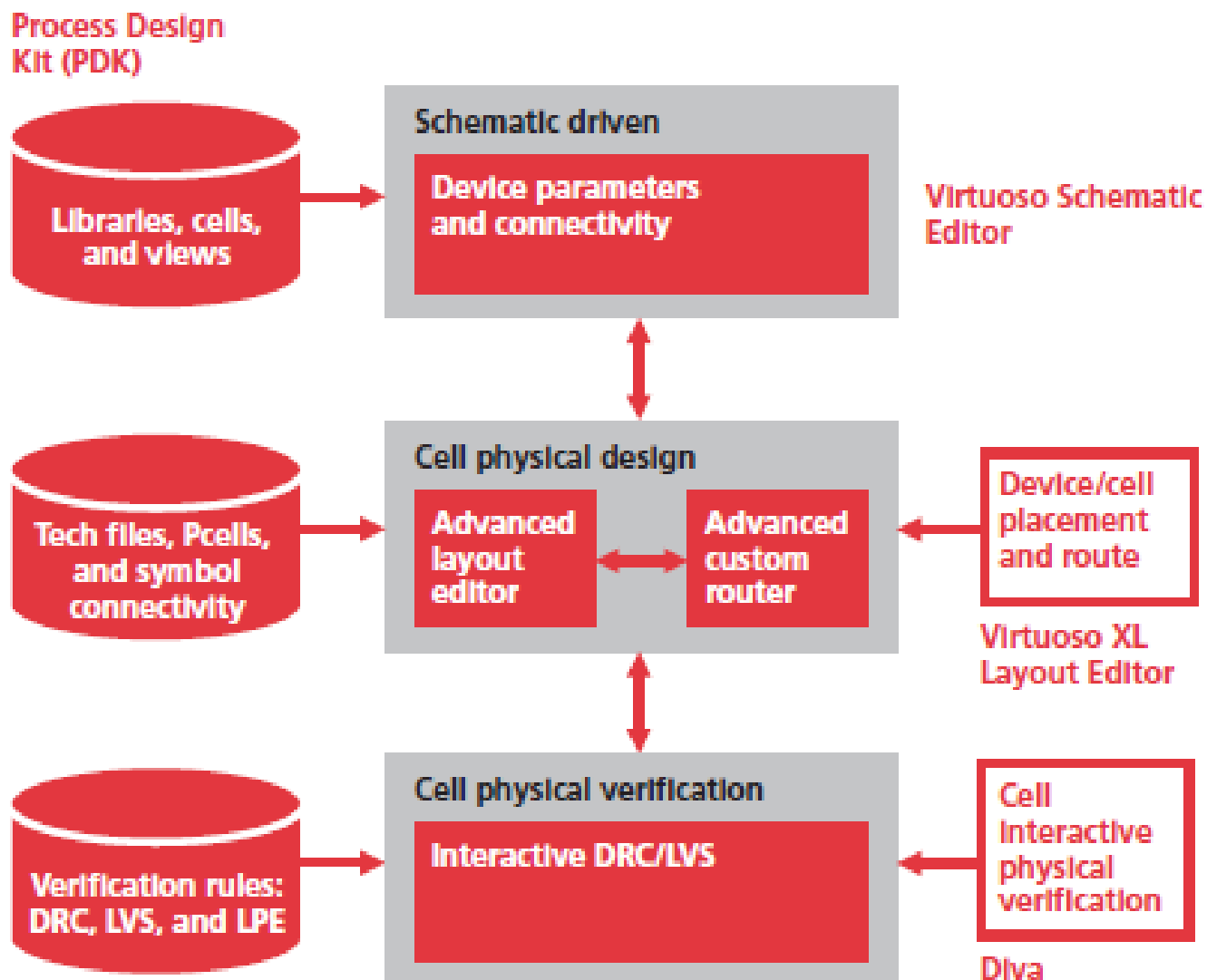
Typical Kit Development Flow



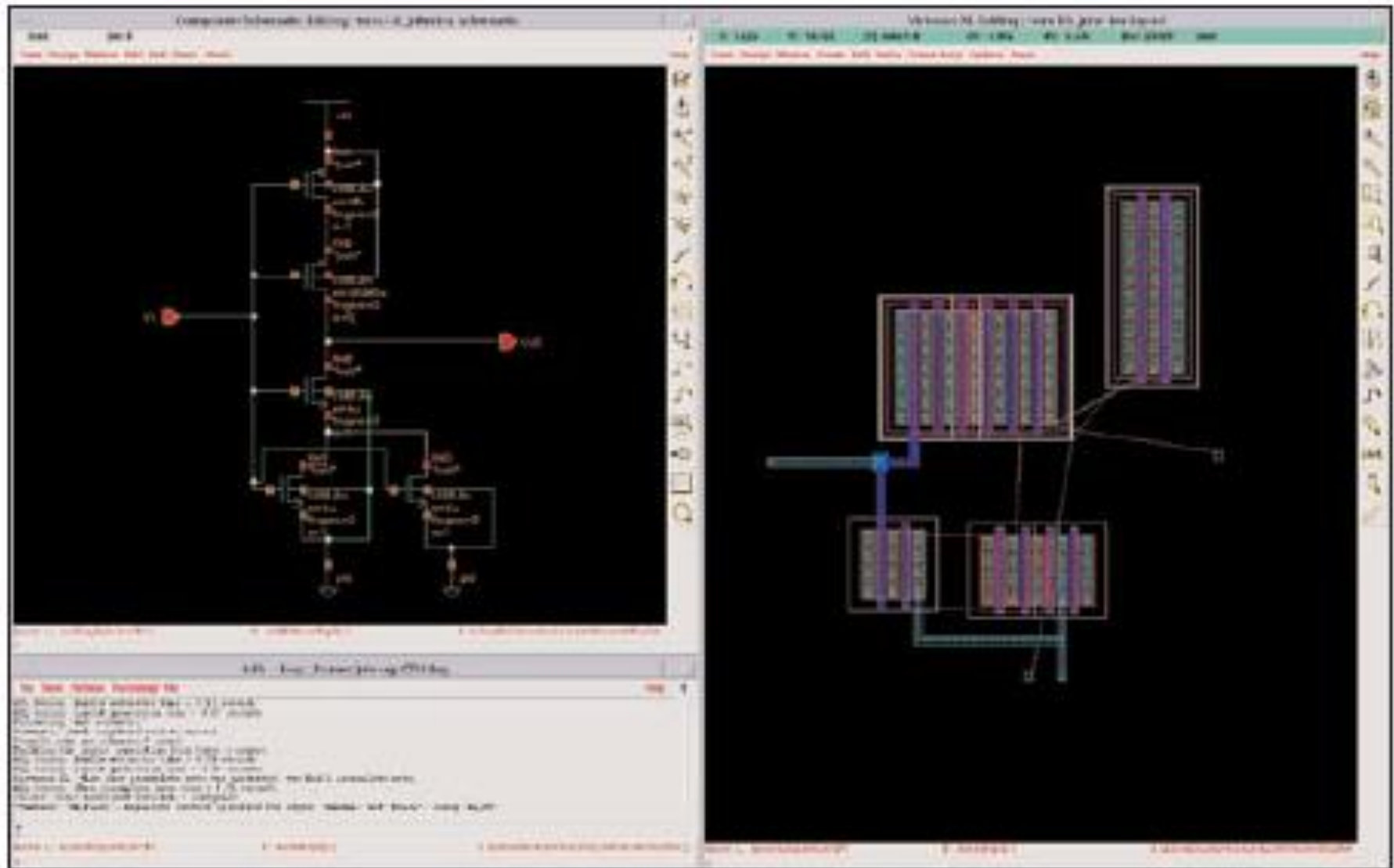
Physical Design Kit (PDK)



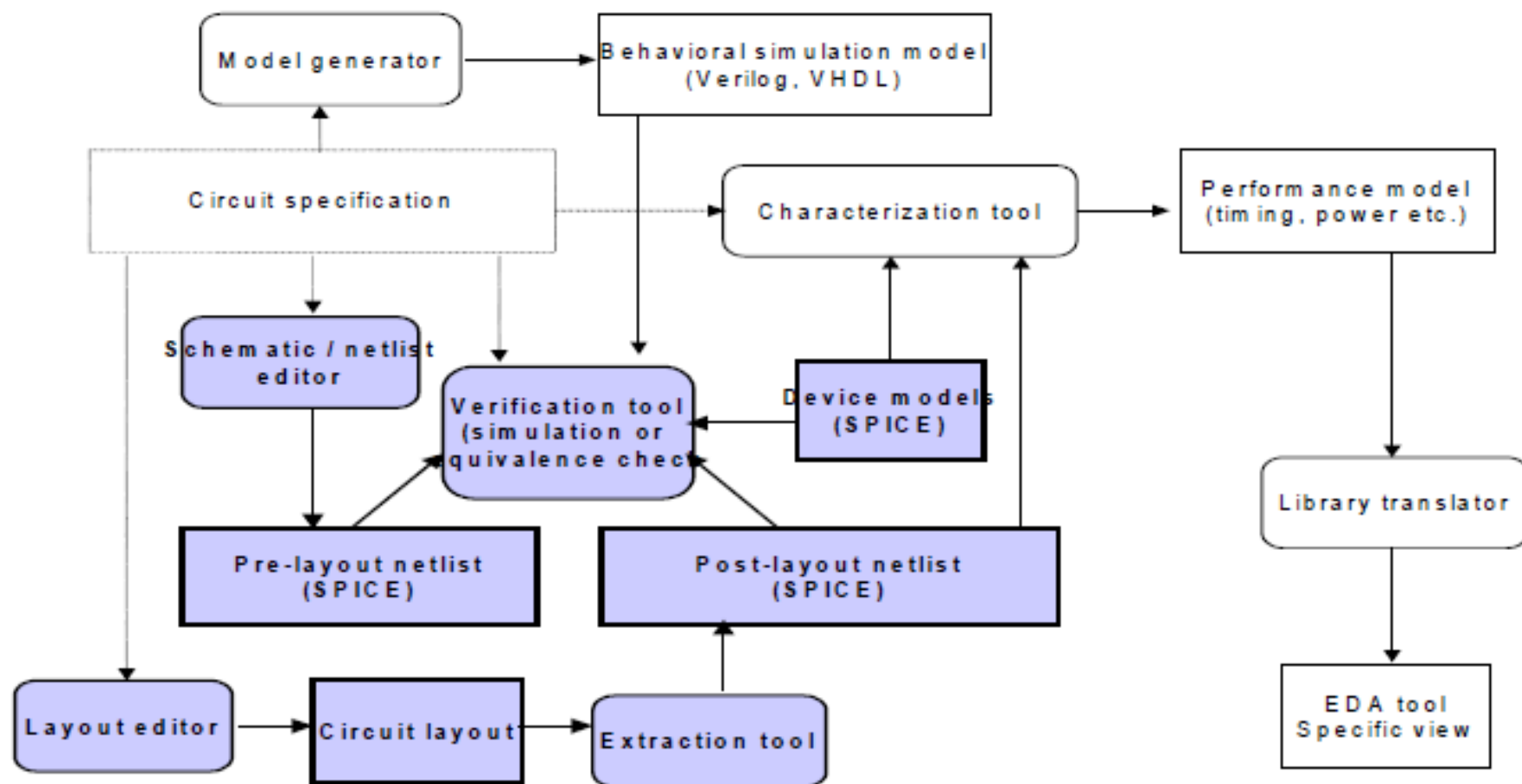
Schematic and Layout



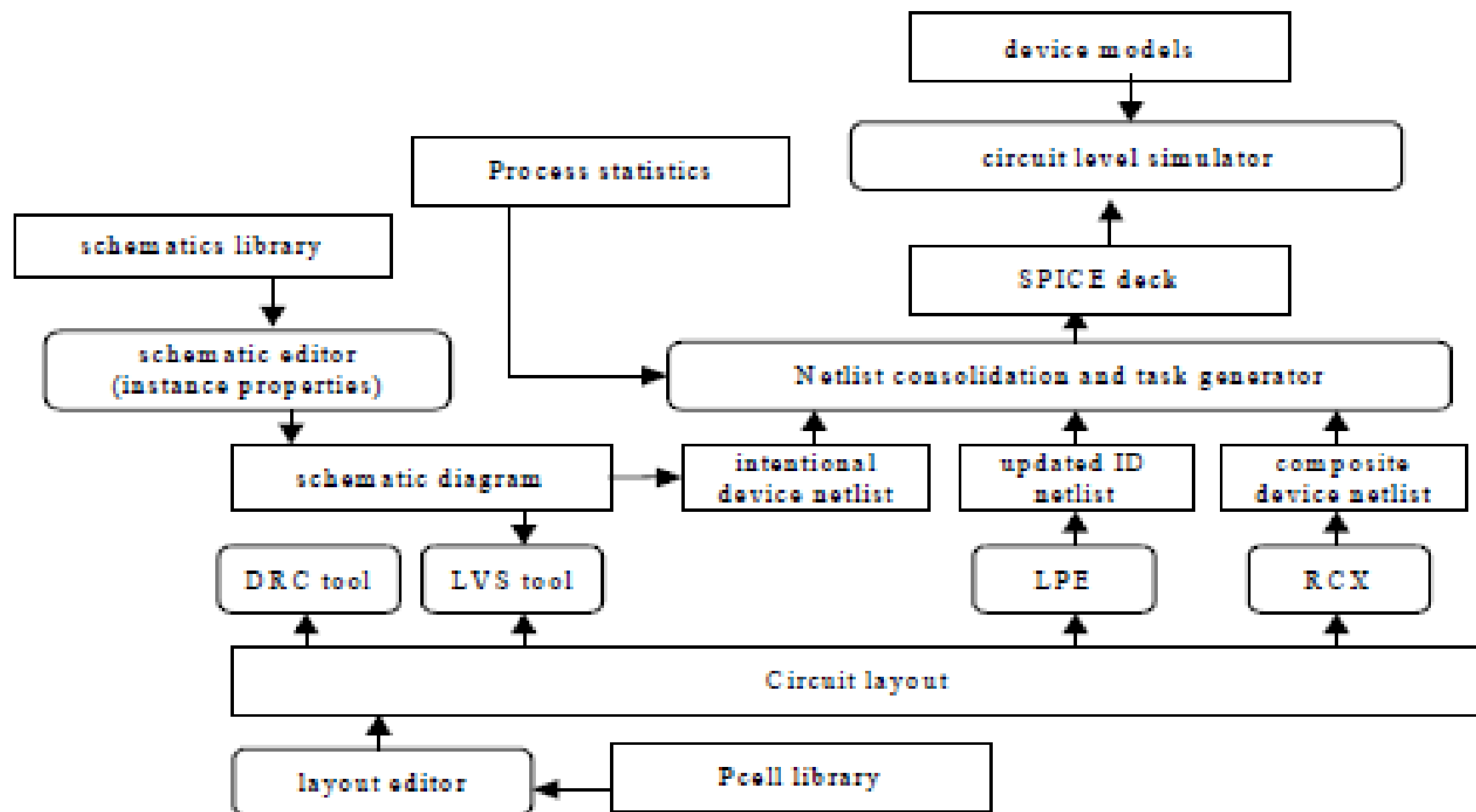
Schematic and Layout



המקום של PDK בתהליך התכנון



רכיבים של PDK



רכיבים של PDK (מושגים)

GENERAL TERMINOLOGY

Term	Meaning / Context
custom circuit design	electrical design, physical implementation, simulation and physical verification of a circuit in which the basic unit of design are individual devices.
device	the basic design element supported by a semiconductor process technology, e.g. transistors, resistors, capacitors, inductors
structure	a building block that is the results from specific semiconductor manufacturing steps, e.g. implants, diffusions, contacts, interconnects, vias, that is used to construct a device. It is possible for structures to also be considered as device.
supported device list	the set of devices that are supported and controlled as part of a process technology definition; supported devices may be intentional or extracted.

רכיבים של PDK (מושגים)

DEVICE TERMINOLOGY

Term	Meaning / Context
intentional device	devices which appear in the schematic diagram; these are the design elements of the circuit design
extracted device	devices which do not appear in the schematic diagram but which are added to the electrical circuit representation (netlist) through analysis of the physical circuit implementation.
main device element	the portion of a device (intentional or extracted) that performs its primary desired function or most significant behavior
parasitic device element	the portion of a device (intentional or extracted) that is present as a by-product of the physical implementation of the main device elements
principal device(s)	the 1 or 2 devices in a technology definition around which the architecture of the process technology is defined, designed, and optimized e.g. NMOS and PMOS devices in a digital CMOS process. Principal devices have the tightest specification tolerances.

רכיבים של PDK (מושגים)

primary device(s)	devices in the supported device list that is built from the structures defined for principal devices with specific additional structures, usually involving one or more dedicated, controllable process steps, e.g. poly resistor built from gate poly with dedicated silicide block mask and implant
secondary device	A device in the supported device list that is built from structures that are defined, designed, and controlled for principal and/or primary devices; these are sometime referred to as "free devices" or "intentional parasitic devices", e.g. using metal interconnect to create an inductor or capacitor. Secondary devices typically have the widest specification tolerances.
device class (category)	general functional category to which a device may be assigned, e.g. MOS, bipolar, resistor, capacitor, inductor, etc.
device type (vertical)	the unique combination of vertical structures that are used to construct and differentiate devices within a class, e.g. poly1 resistor, poly 2 resistor
device style (lateral)	lateral variations within a device type that have the purpose of achieving specified differences in device performance, e.g. modifying lateral layout of EBC of a bipolar transistor to achieve different V_a specifications
device size	special case of style in which the differences in behavior are considered scalable, e.g. modifying gate width in a MOS device

רכיבים של PDK (מושגים)

MODELING TERMINOLOGY	
Term	Meaning / Context
model	the representation of a device used by simulation to represent its function and performance characteristics; models are made from a combination of general behaviors and parameters
model general behavior	portion of a model that represents the generic behavior of a device class and type; can be made from compact model equations, behavioral models, and/or sub-circuits
model process parameters	portion of a model that makes its general behavior and performance related to a specific process technology; e.g. model cards.
instance parameters (allowed design variables)	the properties assigned to an instance of a device that are passed to the simulator and model through the netlist and which describe directly controlled items like device size (e.g. L & W for a MOS device) and calculated items like sidewall capacitance.

רכיבים של PDK (מושגים)

EDA RELATED TERMINOLOGY

Term	Meaning / Context
primitive	the smallest constructor element that is represented within an EDA system; primitives are used to build up device representations
instance	a uniquely identifiable placement of a device within a design
property	information that is associated with and assigned to primitives that specify or modify their behavior
instance property (direct, indirect)	a user-controlled (direct or indirect) parameter associated with an instance; direct properties are assigned by the designer, indirect properties may be calculated or derived from direct properties; sometimes called "designables"

רכיבים של PDK

Element	Description	Foundry	Tools	Design
technology documentation				
device specifications	list of supported devices and their specifications	x		
layout rules	defines coding standard, layer design rules, device-specific design rules, and sample layouts	x		
design guidelines	collection recommended best practices on using the device list and process technology	x		x

רכיבים של PDK

device library				
symbols	schematic representation & properties associated with element in device library		x	
sample layouts (or p-cells)	example of how to correctly layout & code devices; automated versions are p-cells	x		x
placement calculations	calculates indirect parameters (callbacks)	x		
technology file	defines layers and coding standards			
SPICE models	model cards, subcircuits, and behavioral models required to model devices supported by the PDK	x	x	x
physical verification				
DRC decks	checks layout against design rules	x		
LVS deck	checks layout against schematic	x		
LPE deck	updates pre-layout estimates with physical values	x		
RCX deck	adds extracted devices to the circuit representation	x		

בדיקת PDK : Frontend

For each device:

- Device symbol look (size, layout, #pins, pin locations etc.) obeys OKI definitions
- Device allows input of appropriate parameters as defined by standard (e.g. width/length/fingers/multiplicity) for mosfets, w/l/r for resistors etc.)
- Parameter input should allow numeric or string (so user can enter a variable name for the parameter rather than a number, needed for optimization)
- Parameter names are as specified in standard (e.g. "width" or "w"?)
- Device calculates standardized parameters (e.g. ad/as/pd/ps for mosfets) and does it correctly
- Device netlists correct parameters with correct value and syntax for all tools
- Device should not allow entry of incorrect data (e.g. should have checks in place so that user can't enter a device size that disobeys design rules).
- Devices should display tool specific information correctly (e.g. current, voltage from simulators)

בדיקת PDK : Frontend

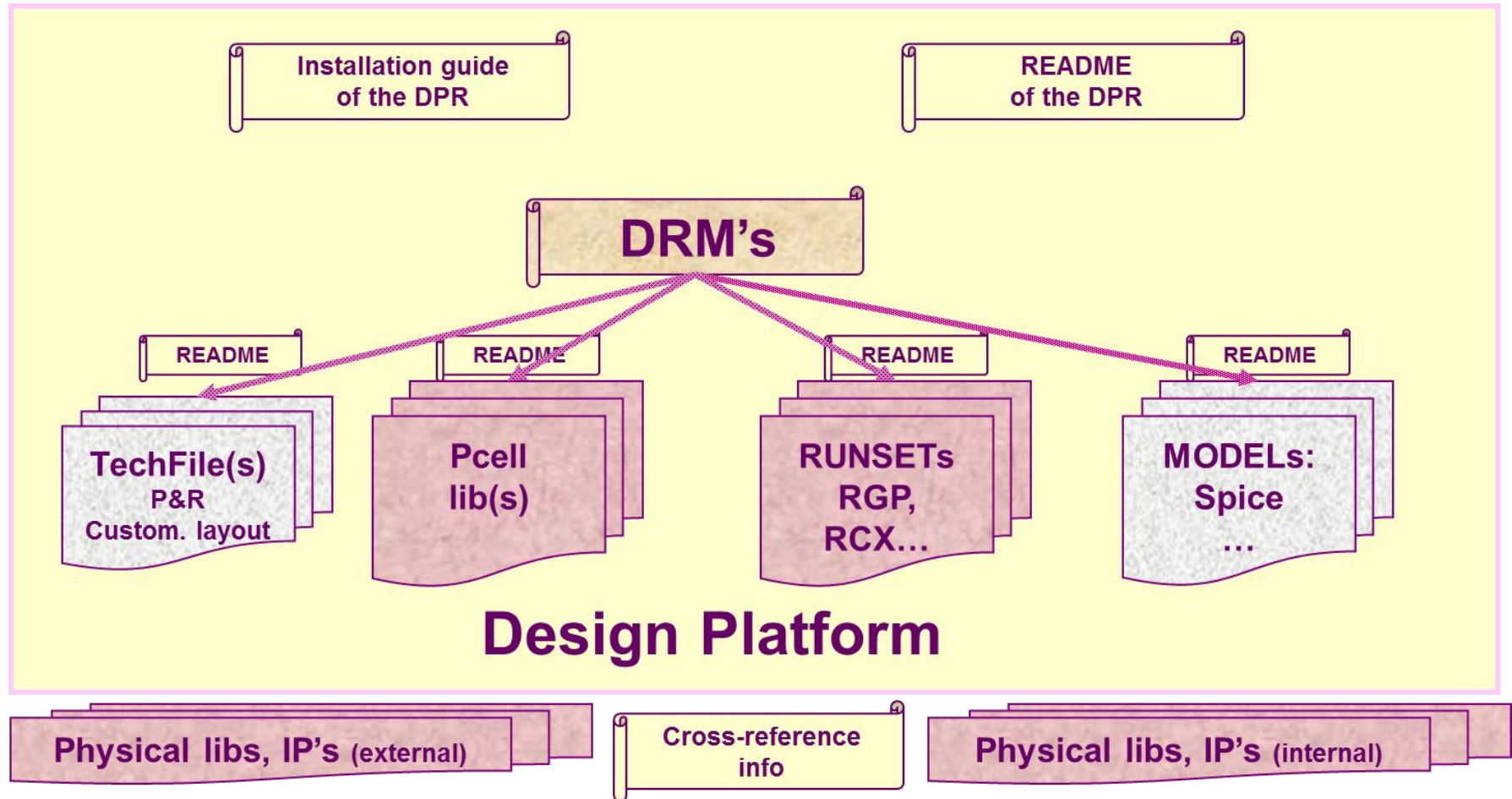
- **Front-end test suite** Create schematic(s) containing one of every device in the PDK instantiated with default values
- Create schematic(s) for each device and for each parameter. Need multiple device instantiations that vary the parameter from min to max with appropriate increments
- Create schematic(s) for each device with parameters set to values used for in layouts for model extraction (compare performance of models to silicon)
- Netlist each test schematic for every supported tool:
 - Verify that all parameters are output correctly (value, syntax etc.)
 - Run each netlist in the tool to make sure no errors are detected and document all tool versions which were tested and results (include replay file to regenerate results if supported by tool) – this will also test syntax correctness of models
 - Include this “golden” netlist with the PDK so user can validate against it

בדיקת PDK : Frontend

– Models Model should contain no syntax errors

- Model is defined for all supported values (e.g. if binning then should have bins covering min to max param ranges) of device model parameters
- Model supports standardized characterization effects (1/f noise, defined threshold ranges, temp coeff, voltage coeff etc.)
- Model documentation includes details of silicon devices upon which model is based (sizes, types, variations etc.)
- Model documentation includes measured data from silicon
- Model is used to generate I/V curves to validate there are no undefined modeling regions. I/V curves are supplied as part of design kit documentation
- Model validation occurs across all tools with correlation between all tools and tested/supported tool versions are documented with model release
- Standardized corner models are supplied and model is tested across all corners

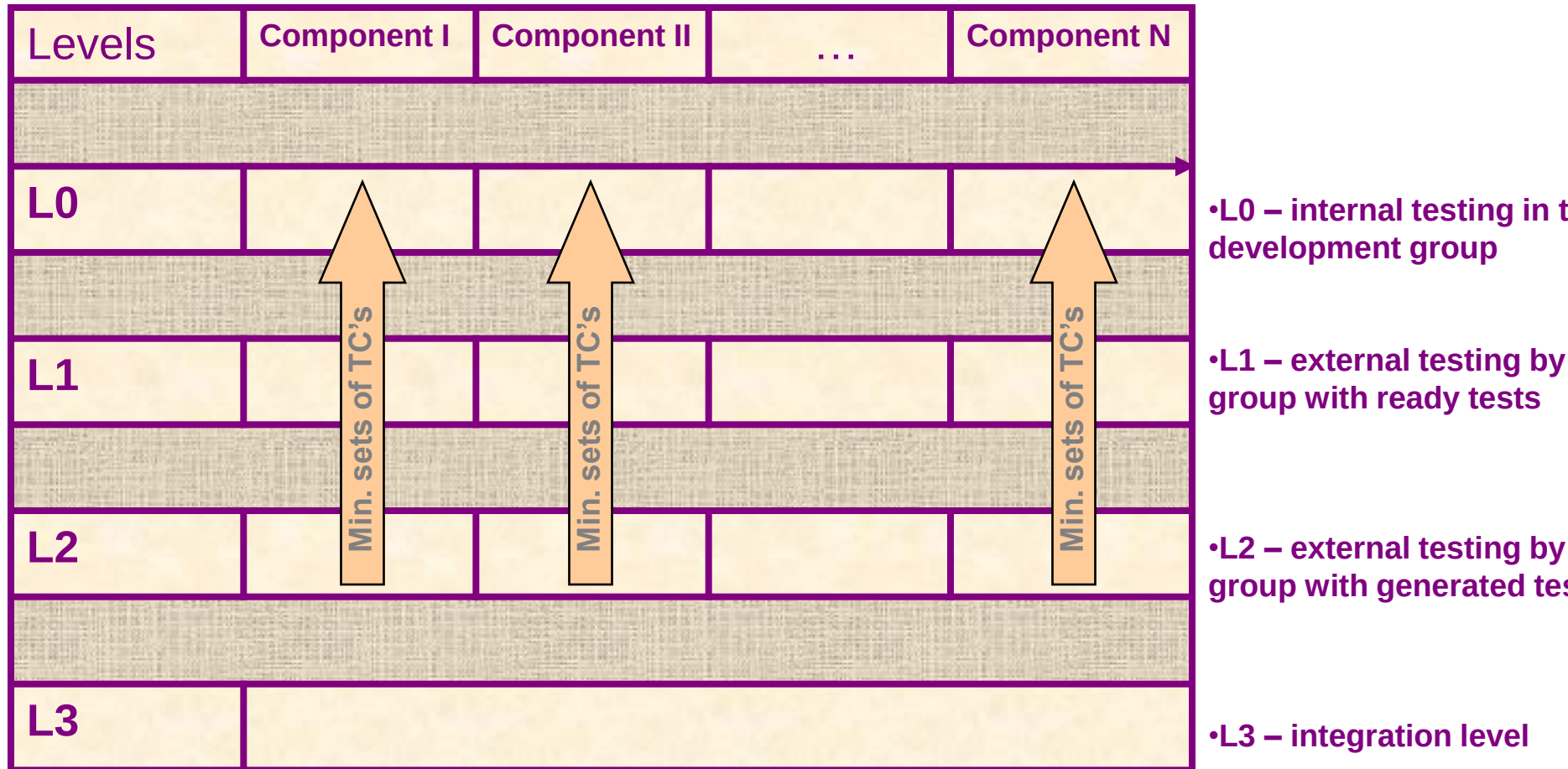
Components of a PDR release



בדיקות

- ✓ Testing done at 4 main levels (+sanity checks):
 1. internal testing in the development teams (L0)
 2. external testing on ready tests (like IP's for DRC), including regressions, new products and reported bugs (L1)
 3. external testing on generated tests (like layout drawn test cases for DRC) (L2)
 4. Integration test (like validation of interface between Pcell lib and models) (L3)
- ✓ All deliverables are TAG'ed: **IFULLQA, IHOTFIX, NO_QA**
- ✓ Customer should not see the TAG's
- ✓ Customer support group should be allowed to see the TAG's

General Testing Concept



בדיקת סיפריות Pcells

- Presence of all legal devices and their views
- Correctness of all names per device
- Correctness of SPICE/LVS terminals and parameters per device
- Correctness of Pcells limits per device
- Correctness of Pcells parameters per device
- Correctness of callbacks per device
- Correctness of all produced netlists for a final set of schemes
- DRC correctness for different combinations of Pcells parameters per device
- LVS correctness for different combinations of Pcells parameters per device
- LVS correctness for a final set of schemes
- Correctness and completeness of tech-Files for each top metal
- Correctness and completeness of sets of symbolic devices for each top metal

בדיקות של Runsets

- Runs on all available IP's and designs
- Runs on manually prepared good and bad test cases
- Runs on all set of gds's and netlists from the corresponding PDK lib (for LVS)

בדיקות מודלים

- Runs on a final set of schemes to validate the correctness of the interface
- Runs on device oriented schemes to validate the correctness of the model
- Runs of mvt